

The University of Texas at Austin

IN-CLASS WORK 6

M339D Intro to Financial Math

September 21, 2026

HEDGING WITH EUROPEAN CALL OPTIONS

PROBLEM 6.1: Sample FM(DM) #13.

Suppose that you short one share of a stock index for 50, and that you also buy a 60-strike European call option that expires in 2 years for 10. Assume the effective annual interest rate is 3%. If the stock index increases to 75 after 2 years, what is the profit on your combined position?

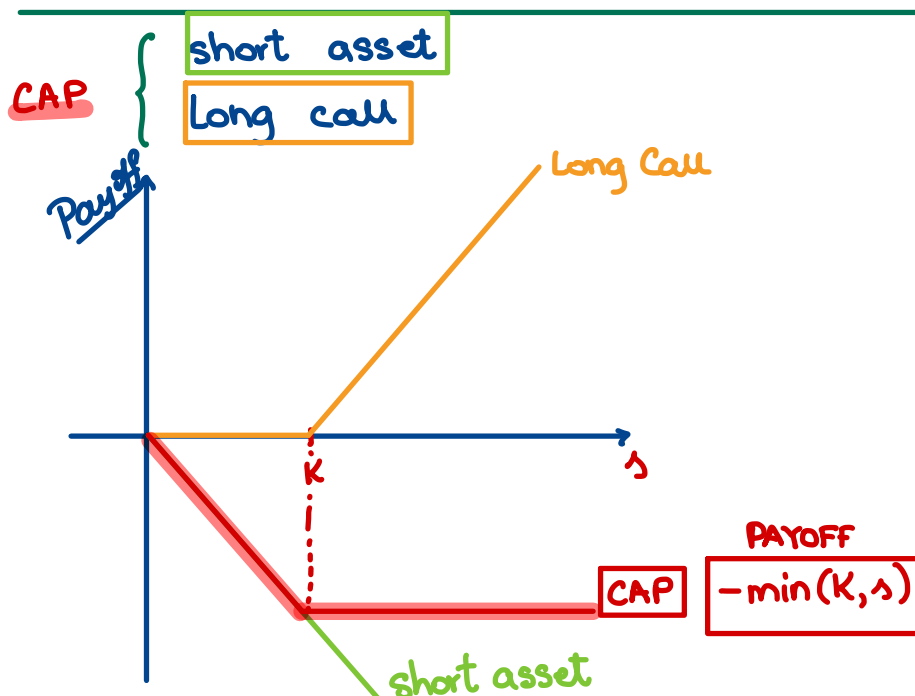
→:

$$\text{Initial Cost: } -50 + 10 = -40$$

$$\text{Payoff: } \underbrace{-75}_{\text{SHORT STOCK}} + \underbrace{(75-60)}_{\text{LONG CALL}} = -60$$

$$\begin{aligned} \text{Profit} &= \text{Payoff} - FV_{0,T}(\text{Init. Cost}) \\ &= -60 - (-40)(1.03)^2 = -17.564 \end{aligned}$$

□



$$\text{Payoff} = -S(T) + (S(T) - K)^+$$

$$= \begin{cases} -\cancel{S(T)} + (\cancel{S(T)} - K) = -K & \text{if } S(T) \geq K \\ -S(T) & \text{if } S(T) < K \end{cases} = -\min(K, S(T))$$

distractors

PROBLEM 6.4: Suppose that you own one share of a stock index, currently priced at \$50, and that you write a \$60-strike European call option on it that expires in 2 years for \$10. Assume the effective annual interest rate is 3%.

You model the stock index price in 2 years as follows:

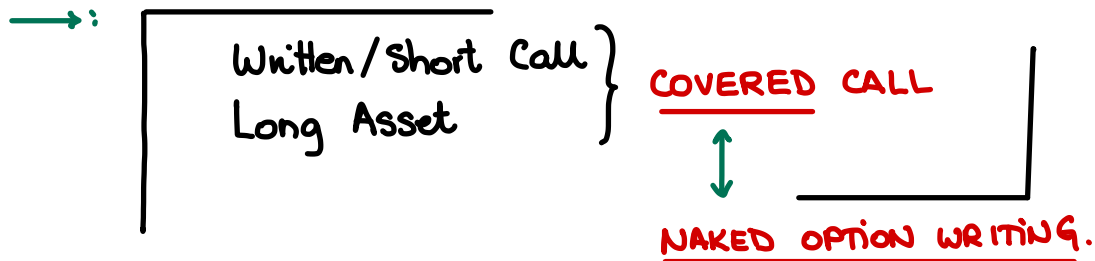
Stock index price in 2 years	Probability
40	0.25
65	0.50
90	0.25

$K \wedge S(T)$

40 w/ prob $\frac{1}{4}$

60 } w/ prob $\frac{3}{4}$

Calculate the expected payoff of your covered call position.



So, our covered call is the short version of the cap.

⇒ Its payoff: $\min(K, S(T))$

$$\mathbb{E}[\min(K, S(T))] = \cancel{X}$$

$$= 40 \cdot \left(\frac{1}{4}\right) + 60 \cdot \frac{3}{4} = 10 + 45 = 55$$

